

DUNKIRK LH, NY, USA

WMO: 994250

Lat: 42.4939N

Lon: 79.3540W

Elev: 183

StdP: 99.15

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.2	(c) -12.0	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 25.3	(k) 0.9	(l) 22.6	(m) 0.2	(n) 6.9	(o) 220	(p) 0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 5.5	(c) 27.7	(d) N/A	(e) 26.6	(f) N/A	(g) 25.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.0	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 19.4	(b) 16.2	(c) 14.0	(e) -16.2	(f) 30.8	(g) 3.6	(h) 1.2	(i) -18.8	(j) 31.6	(k) -20.8	(l) 32.3	(m) -22.8	(n) 33.0	(o) -25.4	(p) 33.8		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-2.3	-2.0	1.1	7.1	13.2	18.9	21.6	21.2	18.0	12.2	6.5	1.4
	DBStd	9.66	6.02	5.50	5.44	4.72	4.26	3.45	2.48	2.38	3.43	4.07	4.70	4.77
	HDD10.0	1552	384	338	282	113	16	0	0	0	0	23	126	271
	HDD18.3	3417	641	571	535	337	168	34	3	4	47	196	356	526
	CDD10.0	1478	2	1	6	27	114	267	359	348	239	90	21	3
	CDD18.3	299	0	0	0	1	9	51	104	94	36	4	0	0
	CDH23.3	1062	0	0	1	6	27	160	403	331	123	11	0	0
	CDH26.7	88	0	0	0	1	3	16	31	24	14	0	0	0
Wind	WSAvg	5.7	7.6	6.9	6.3	6.3	5.3	4.2	4.1	3.9	4.4	5.6	6.8	7.2
Precipitation	PrecAvg	1070	66	52	67	86	86	95	96	98	124	104	112	85
	PrecMax	1476	143	132	138	163	173	249	234	229	301	188	308	179
	PrecMin	798	24	11	14	23	24	28	28	22	54	24	37	33
	PrecStd	160	29	28	28	34	39	46	49	42	56	35	50	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	16.5	19.8	24.1	26.3	28.6	29.0	28.8	28.5	24.8	20.0	17.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	13.0	11.6	15.9	20.4	23.5	26.5	27.5	27.2	26.2	22.3	17.7	12.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	8.8	7.9	12.6	17.4	21.7	25.2	26.5	26.2	24.7	20.2	15.7	10.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	5.4	4.7	8.7	14.6	19.8	23.9	25.5	25.2	23.2	18.2	13.5	7.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	6.0	6.2	6.5	7.2	6.7	5.8	5.5	5.7	6.3	5.9	5.5	5.2
		MCDBR	9.3	10.5	12.1	12.9	9.6	7.5	6.2	6.4	7.6	8.3	8.4	8.3
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.306	0.342	0.337	0.375	0.405	0.437	0.439	0.429	0.380	0.362	0.316	0.304
	taud		2.393	2.257	2.366	2.321	2.285	2.232	2.249	2.269	2.377	2.407	2.506	2.459
	Ebn at Noon		832	847	906	892	870	840	833	830	850	820	821	805
	Edh at Noon		79	107	108	122	130	138	134	128	106	92	71	69
All-Sky Solar Radiation	RadAvg		1.28	2.12	3.17	4.41	5.43	5.91	5.96	5.29	4.16	2.54	1.55	0.98
	RadStd		0.12	0.21	0.41	0.47	0.55	0.44	0.46	0.30	0.39	0.26	0.24	0.14

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page